

Hardware for Deep Learning: An investigation for Current Options

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Content synopsis

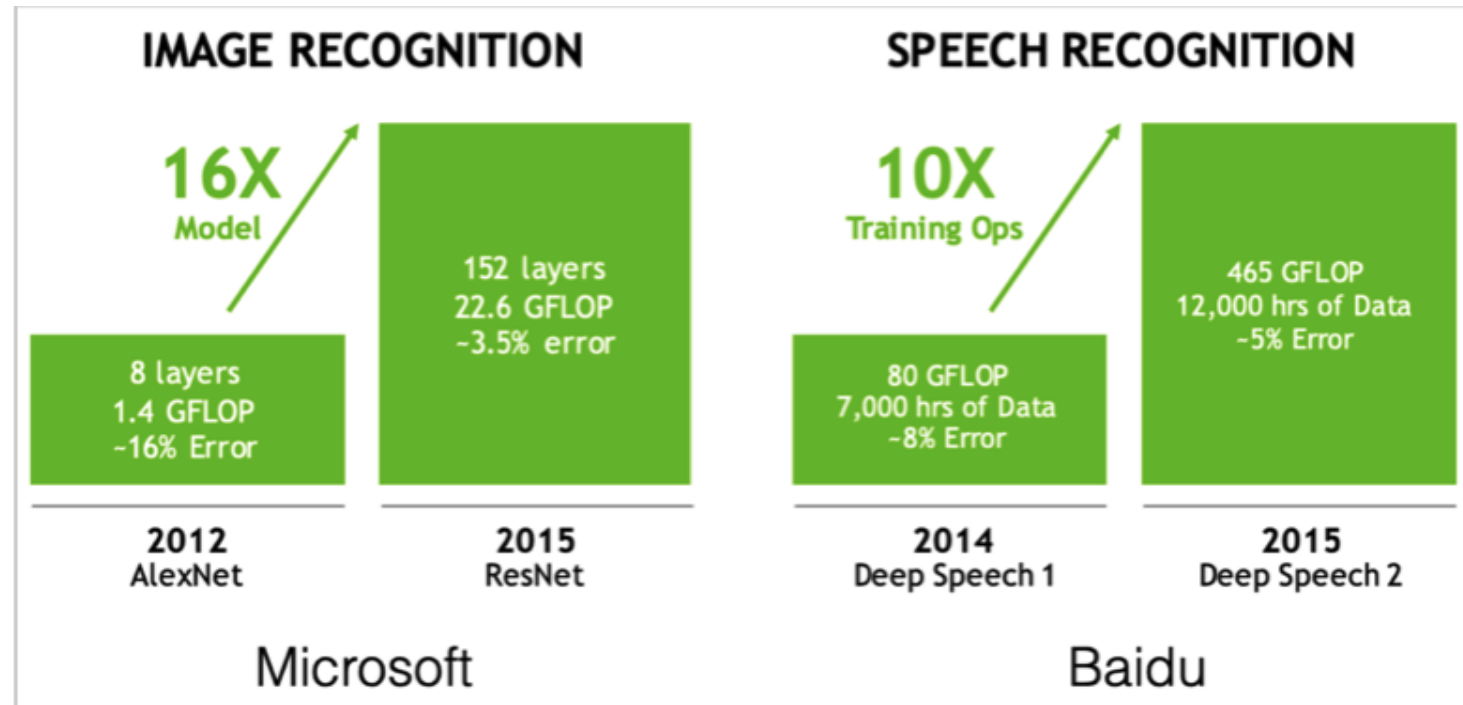
- An introduction to current challenges of hardware for DL models
- Mainstream HW options & benchmark testings
- Summarization

An introduction to current challenges of hardware for DL models

- DL appears in many applications
 - computer vision(auto driving, etc.), NLP, LLM
 - Very deep networks -> large computational power
- Compute-intensive models invoke below challenges:
 - 1.The size of models and data for training & inference gets larger and larger.
 - 2. energy consumption
 - 3. memory bound when loading and storing large data

Size growth of models & data for AI

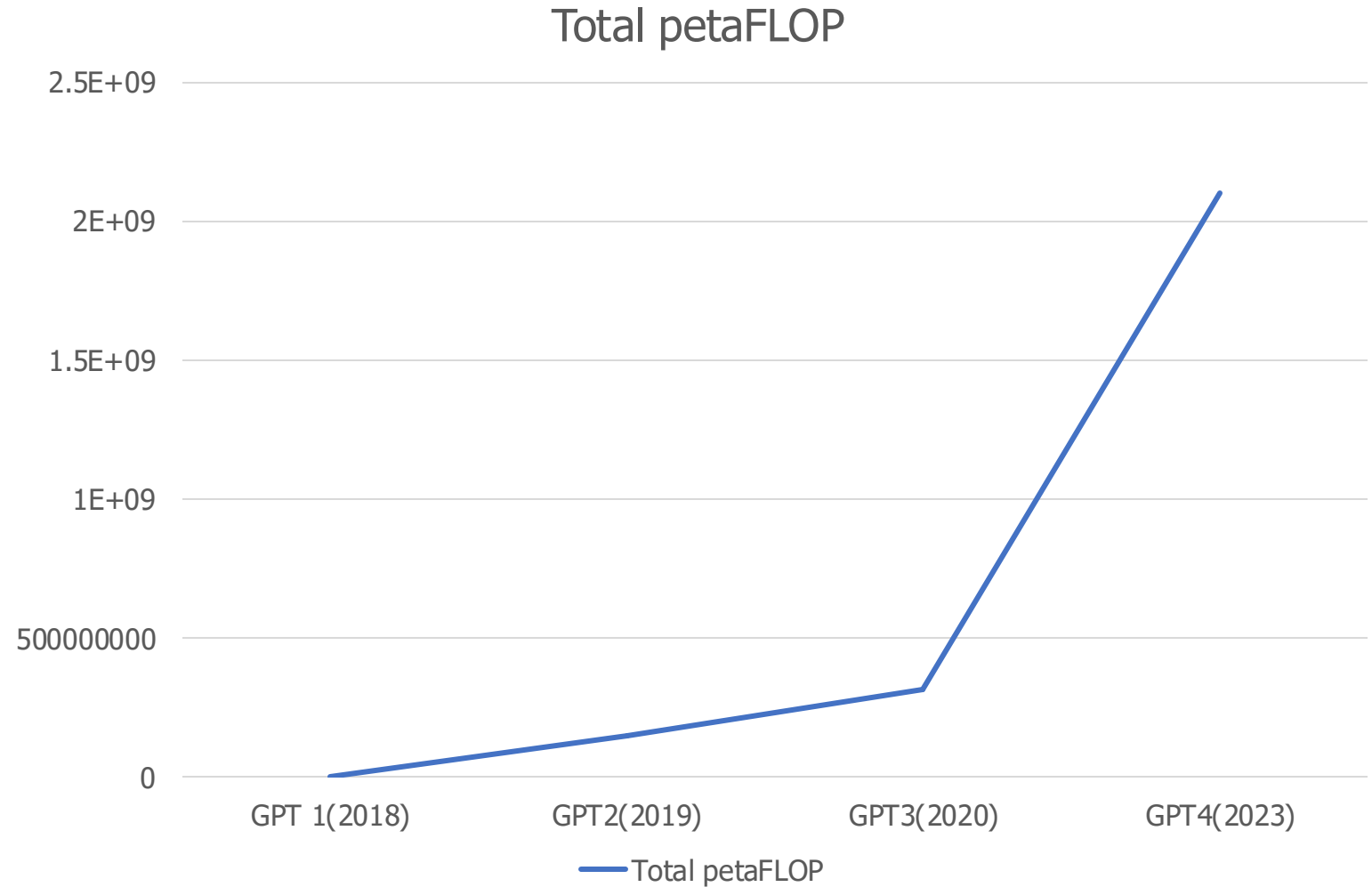
- As the size of model & datasets and DNN model for deep learning increases, we see the dramatically increase of :
 - Training computation(gauged in total FLOP)
 - Training time



Size growth of models & data for AI(Cont.)

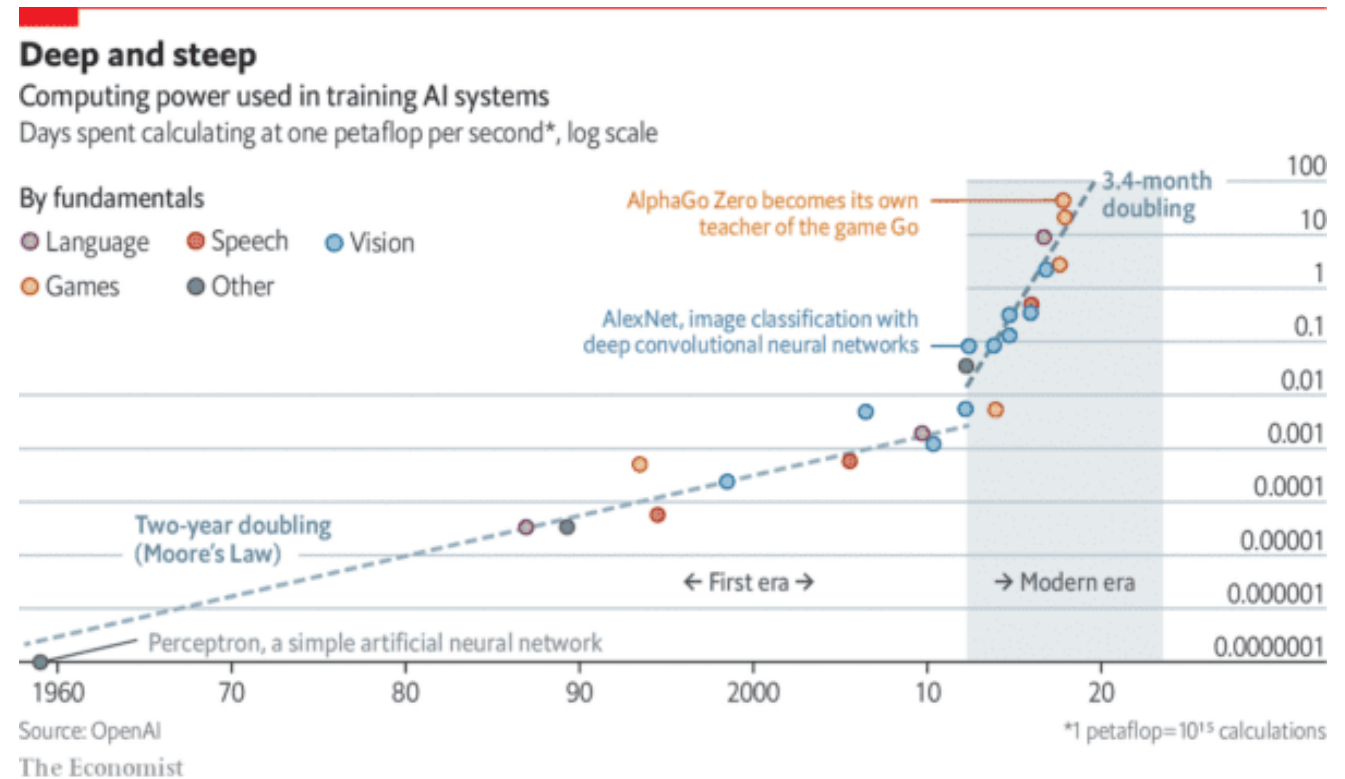
- For LLM:

- GPT1:17,600 petaFLOP
- GPT4:21 billion petaFLOP



Energy consumption

- 1. Computer power used in training AI systems has exponentially increased in the era of deep learning[1]
 - An amazing fact: inference **can exceed** the cost of training in pervasive systems[2]
- [1]<https://www.numenta.com/blog/2022/05/24/ai-is-harming-our-planet/>
- [2]https://www.sciencedirect.com/science/article/pii/S2210537923000124?_cf_chl_tk=5Or2yZTf96pOd_9yGc9Sn1tSTtJyoElQvXzq.qTPaGA-1712373631-0.0.1.1-1386

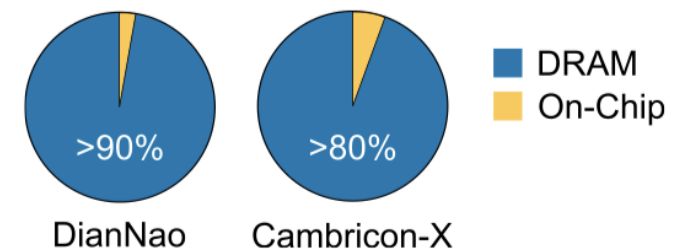


Energy consumption(Conti.)

- 2.Power-limited scenarios: e.g. Internet-of-Things (IoT) applications
 - Some edge nodes is subjected to strict power constrains
 - Not feasible to integrate a high-end GPU
- Thus, there has been a pressure to build DNNs that are less resource demanding
 - Choose more energy efficient options

Memory bandwidth

- Bottleneck of memory fetch & store for inputs & results
- Every MAC requires:
 - Read from: input pixel, weight and partial sum
 - Write: update partial sum
- a DRAM access has an energy cost of ~ 2 orders of magnitude higher than a MAC operation[1]
- Optimization of both HW & SW

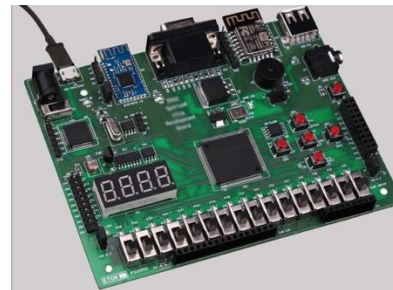
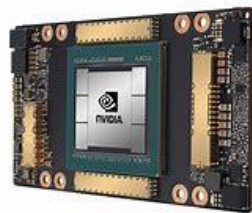


[1]M. Horowitz. Energy Table for 45 nm Process. Stanford VLSI Wiki. Accessed: Nov. 26, 2020. [Online].
Available: <https://sites.google.com/site/seecproject>

FIGURE 27. Energy breakdown of two state-of-the-art DNNs accelerators.

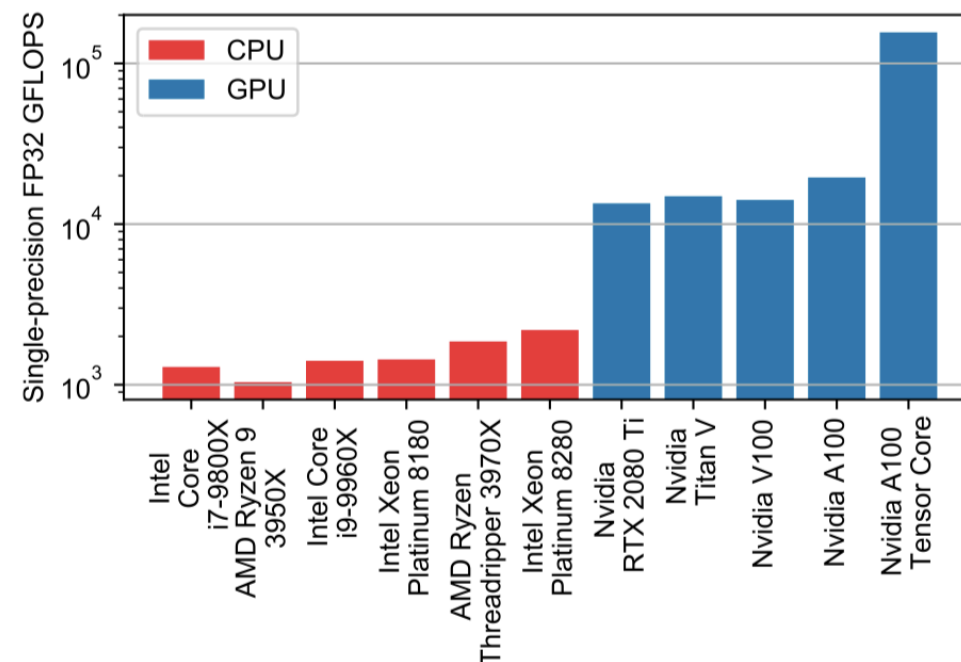
SO, what kind of HW is a suitable option?

- The fast innovation of AI in deep learning and the challenges in research and application makes us seek for suitable options:
 - Meet the needs for specific tasks, scenarios in bearable budget.
- A general comparison of hardware options: CPU, GPU, FPGA, ASIC with 2 Benchmark testing
 - Just rough testing, since there are many factors affecting performance comparison



CPU

- Architecture
 - designed for serial operations
 - less cores & more cache memory
 - Lower FLOPS & FLOPS/WATT performance
- Thus, often the least used for DNNs training or inference
- But, nowadays can be realized as vector processors(e.g. Intel's Skylake-X CPUs)
 - Adding Vector Neural Network Instructions to ISA: Intel's AVX-512 to accelerate CNNs
- Scenario: data preprocessing(image processing)

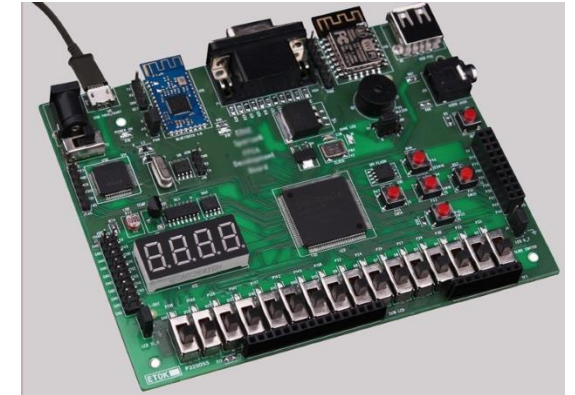


GPU

- Architecture
 - up to thousands of small cores(Titan X: 3500 cores delivering 11 tera-flops)
 - designed for parallel computation: Single-Instruction-Multiple-Threads
 - higher throughput
 - Tensor Core on A100: optimized for large matrix operation
- SW support(Nvidia as an example)
 - CUDA: HW driver, CUDA API & Math libraries
 - cuDNN library provides optimization for DNNs
 - Better support for DL frameworks(Pytorch/Tensflow):
 - saves the end user from implementing parallel code and understanding the architecture of their chip
- scenario:best option for training
 - can accelerate large num of MAC

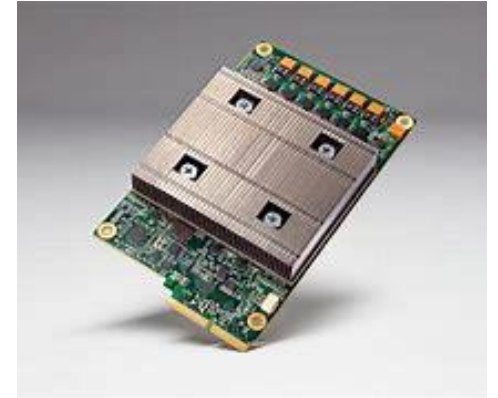
FPGA

- Field Programmable Gate Arrays: a sea of floating logic gates
- Architecture
 - **Programmable:** use Verilog or VHDL
 - adaptable architecture, custom data types(compared to GPU)
 - comparatively lower power consumption
- Disadvantage
 - Need knowledge of hardware design
 - Adaptation to mainstream ML framework written in Python, C++
 - Lack of generalization for large-volume scenario
- Scenarios:
 - - embedded applications, video streaming & other real-time applications
 - - safty-critical operations(advanced driver assistant systems)



ASIC

- application-specific integrated circuits
 - E.g.: Google's TPU, Intel's Nervana & Movidius
- Advantage:
 - highest performing, more energy-efficient than FPGA
- Drawbacks
 - Fixed logic -> least flexible, cannot be reprogrammed
- Scenario:
 - used for specific purpose(data center)



Benchmark Test 1

- Can CPU perform matrix multiplication faster?
- CPU: AMD Ryzen 5500(support AVX2, AVX256)
- DGEMM algorithm matrix-matrix multiplication
 - Basic(unoptimized)
 - Basic & blocked
 - AVX with 256-bit
 - AVX256 & loop unrolling
 - AVX256 & loop unrolling & blocked

```
haibo@DESKTOP-ED1N04Q:~/project/matrix-matrix-multiply/build/src$ ./dgemm
      dgemm_basic: elapsed-time=    331830
dgemm_basic_blocked: elapsed-time=   172974      speed-up=    1.91838
      dgemm_avx256: elapsed-time=    82595      speed-up=    4.01756
dgemm_unrolled: elapsed-time=    27877      speed-up=   11.9034
dgemm_blocked: elapsed-time=    27551      speed-up=   12.0442
haibo@DESKTOP-ED1N04Q:~/project/matrix-matrix-multiply/build/src$
```

- Testing result: elapsed-time & speed-up
 - AVX with 256-bit gets 4 times speed-up
 - AVX with 256-bit & loop unrolling & blocked gets **12 times speed-up!**
- The improvement from both hardware structure & instruction order optimization

Benchmark Test 2

- Training time comparison between CPU vs. GPU T4 vs. TPU on free Colab(Google Cloud)
- Handwritten digits recognition(MNIST dataset)
- Basic CNN model(0.41 million parameters), 10 Epoch, 1024 batch size
- Testing result^[1]:

	CPU	GPU T4	TPU
Epoch Time	76s	2s	2.2s
Step Time	1s	33.7ms	34.8ms
Accuracy	0.9731	0.9729	0.9745

- [1]:Since device may take longer time to compile the model and distribute data in Epoch 1. Only reported the time between Epoch 2-10.

Summary 1

	Architecture feature	Generalization	Energy consumption	Floating-point processing
CPU	Sequential computation; vector computation unit	good	various	strong
GPU	High parallelism & memory bandwidth	good	various	strong
FPGA	Programmable gate arrays	kind of flexible; Reconfigurable; Low latency & high bandwidth	Better than GPU	Don't offer peak floating point perf.
ASIC	Specialized design Highest performance	Least flexible; Huge investment	Better than GPU	

Summary2: application scenario

- CPU: Generalized computation
 - Control workflows (loading weights from GPU memory)
 - model inference
- GPU: Graphic & image processing, neural network training
 - training: Forward & back propagation: batch hundreds of training inputs (image classification or speech recognition) & operate on them
- FPGA: Specialized scenario with low-volume requirement
 - Wireless Communications, embedded devices, datacenter, .., specialized acceleration tasks
 - Training on GPU, but deployed on an FPGA
- ASIC: high-volume requirement, high-degree customization
 - Google's TPU for datacenter

Summary3: trending for AI hardware

- For HW designer & manufacturer:
 - Design architecture for matrix-matrix multiply & convolution operations speed-up(AI PC)
 - Add data/memory paths specialized to machine learning workloads
 - Develop application-specific functional units
- For AI model user & developer:
 - Choose suitable hardware device for model training and inferencing: time & budget
 - To make ML frameworks more adaptable to different kinds of devices(FPGA etc.)
 - Use Apache TVM(<https://tvm.apache.org/>) for deployment & optimization

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